

Formulating Absolute Value Linear Equations - Practice

Algebra II | Unit 2 | Lesson 4 Practice | TEK A2.6.D

Objective: I can formulate an absolute value linear equation from a real-world description using the $|x - \text{center}| = \text{margin}$ template.

Section 1 — Recall

1. A target weight is 25 lb, with a tolerance of 2 lb. Formulate an equation - do not solve.

2. A length must be within 0.3 cm of 15 cm. Formulate an equation - do not solve.

Section 2 — Apply

3. A machine's temperature must stay within 5% of the ideal setting of 200°F. Formulate an equation - do not solve.

4. A runner's pace must be within 30 seconds of a target time of 360 seconds per mile. Formulate an equation - do not solve.

Section 3 — Challenge

5. A company ships boxes that should weigh 20 lb. Any box more than 1.5 lb heavier or lighter gets flagged. Formulate the equation for the boundary weight where flagging begins.

6. The equation $|x - 15| = 3$ came from a real-world scenario. Describe a possible scenario, naming what 15 and 3 stand for.

Section 4 — Explain It

7. Explain what "center" and "margin" mean in a real-world $|x - \text{center}| = \text{margin}$ equation, and why the equation always takes that form.
